



When you need
to be sure
服务机器人相关认证浅析



目录

01 关于我们

02 认证方案

03 认证流程&案例

概览

全球化服务网络, 本土化专业能力



公认的
行业领导者



99,600名
员工



2,600 个
分支机构和实验室



5 大
事业群

无论您身在何处, 从事何种行业, SGS遍布全球的专家都能提供专业解决方案, 助力业务进程更快捷、更简便、更高效。

我们的里程碑

SGS由 Henri Goldstuck 创建于法国鲁昂

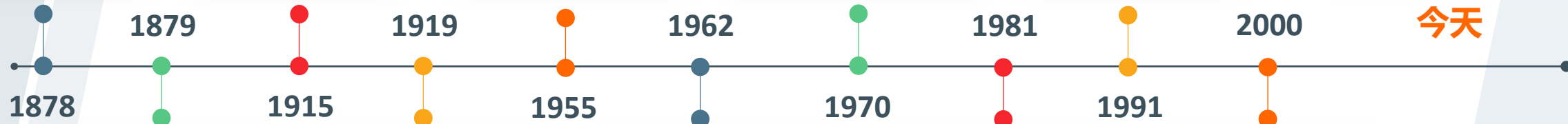
总部从巴黎迁至瑞士日内瓦

SGS进入工业领域，开始工业机械和工业产品的检验业务

70 年代，SGS开始为石油和天然气行业提供环境咨询及修复服务

亚洲经济的迅猛增长促使我们进入中国市场

SGS的业务遍布全球的各行各业，我们的99,600多位员工分布在2,600个分支机构和实验室



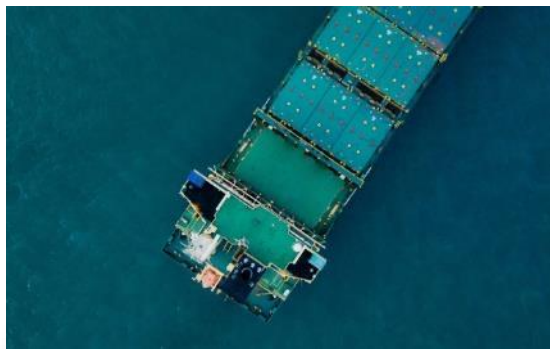
我们的商业原则



帮助客户提升竞争力



提升质量
降低风险



加速入市



改善安全、效率、
生产力和可持续性



建立信任

SGS在中国

通标标准技术服务有限公司（SGS中国）成立于1991年，总部位于北京，是SGS集团和隶属于国家市场监督管理总局系统的中国标准科技集团共同组建的合资公司

中国境内首家获得CNAS（中国合格评定国家认可委员会）ISO 17020认可的外资检验机构，也是国内获CNAS认可范围最广的外资认证机构

资质完备的实验室网络，获得包括CNAS, CMA, IECEE, GS, DAKKS, UKAS, HOKLAS, KFDA, JPMAS, ISTA, CCC, cGMP等多家国际权威机构颁发的实验室认可



电子电气实验室，中国

第三方测试、检验和认证行业的标杆企业



16,000名
专业员工



连续15年
入选中国最佳人力资
源典范企业100强



100+
分支机构

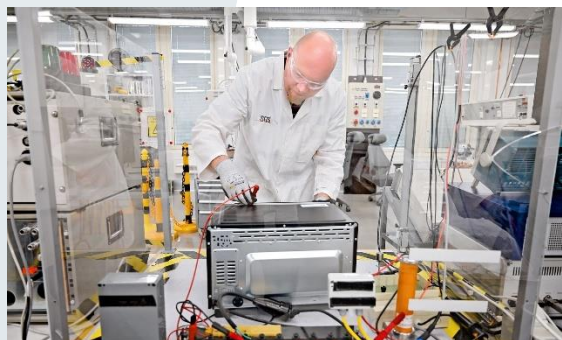


200+
实验室



五大事业群

测试和检验



互联与产品



营养与健康



工业与环境



自然资源

认证



管理与保证

测试和检验:

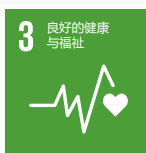
互联与产品

我们保障互联、消费品、商用机械及贸易的质量、安全和可持续性

互联与产品涵盖:

- › 汽车服务
- › 电子电气机械服务
- › 半导体及可靠性服务
- › 时尚与服饰科技服务
- › 轻工产品服务
- › 科技电子产品限用物质测试服务
- › 产品检验及现场服务
- › 消防科技服务
- › 贸易保障服务

可持续发展目标的影响



电子电气实验室, 芬兰



C&P EE commercial

商用机械产品领域一站式合规服务

电子电气产品涵盖：

- › 工业/商业机械产品认证
- › 工业功能安全评估
- › 电能表/计量产品认证
- › 防爆产品认证
- › 半导体制程设备认证



商用/工业用产品服务-产品类别

商用/工业机械产品领域一站式合规服务

- › 压缩机/冷却系统
- › 风力涡轮设备/鼓风机/泵
- › 起重机/升降设备
- › 建筑机械/工业卡车
- › 玻璃机械
- › 工程机械
- › 金属加工/成型设备
- › 铸造设备
- › 连续输送设备
- › 食品加工机械
- › 服务机器人
- › AGV
- › 工业/协作机器人
- › 气动工具
- › 皮革及鞋类加工机械
- › 低压开关和控制设备
- › 包装/印刷/造纸设备
- › 塑料/橡胶加工
- › 缝纫/纺织机械
- › 木材加工机械
- › 生产线/物流线
- › 机电设备
- › 安全部件/光幕/控制器
- › 激光/半导体制程设备
- › 非标锂电产线
- › 手持式/便携式电动工具



培训



企业定制化培训

由专业专家设计，可帮助组织和个人提高个人能力和技能，这是可持续业务发展和竞争优势的核心促进因素。



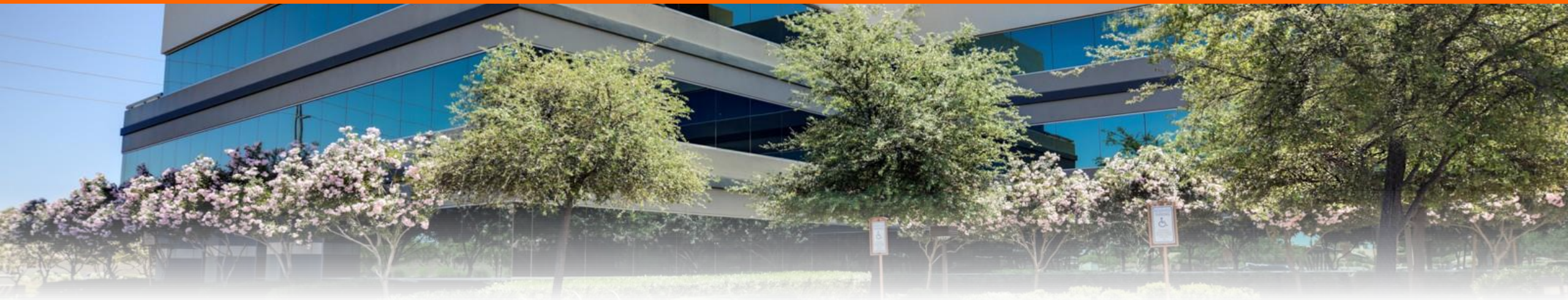
法规培训

欧盟CE、北美NRTL等多地市场准入，风险评估（ISO 12100）等



产品标准培训

C类产品标准，例如机器人，AGV，SEMI等特殊产品标准的针对性培训



具身机器人合规思路

1

ISO 13482

服务类型机器人

2

ISO CD 25785-1

工业人形机器人

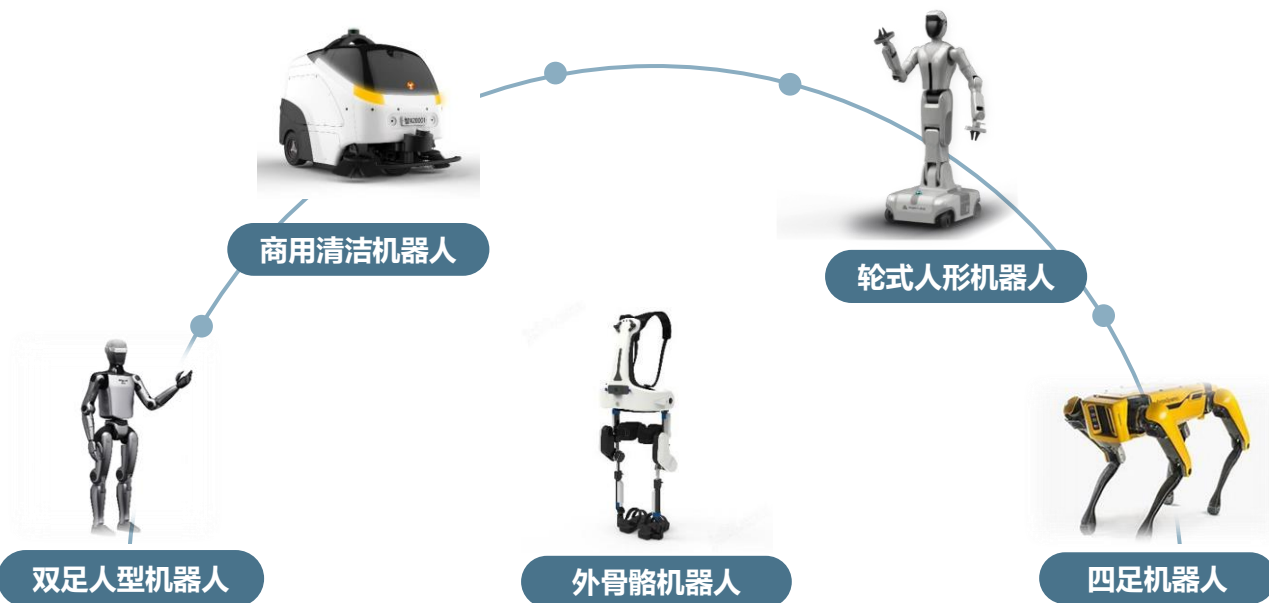
3

IEC 63327

商用清洁机器人- Particular requirements



典型机器人场景



具身机器人合规思路





一般需求：机器人评估

- 环境模拟与耐久性
- 电气测量及损坏与失效分析
- 电磁兼容性 (EMC)
- 电池测试，包括电池分析
- 化学品与火灾测试
- 无线与连接测试 (OTA)
- 功能安全——包括半导体
- 网络安全
- SOTIF (机器人学)
- 人工智能 (AI)

Wiring, connectors, sensors, actuators, electronic control units, batteries, communications, entertainment systems, lighting, Telematics Control Units, V2X OBU / RSU, Vehicle Antenna Performance, Radars, Lidars

五层合规考量

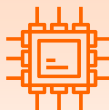
从法规到芯片，层层递进，构建完整合规供应链

0 | Layer 1: Legal and Regulatory Framework 

0 | Layer 2: Governance and Management System 

0 | Layer 3: Machinery and Robotics Safety 

0 | Layer 4: E/E System Level development 

0 | Layer 5: Hardware and Software Level (including semiconductors) 



Layer 1 – Legal & Regulatory Framework (Top-Level Obligations)



Area	Relevant Framework
AI Regulation (EU)	EU AI Act (High-Risk AI if safety-relevant control)
Product Safety	Machinery Regulation 2006/42/EC (new 2023/1230)
Product Liability	EU Product Liability Directive
Workplace Safety	Occupational Safety Directives
Cybersecurity	Cyber resilience act (CRA/EU)



This layer defines **mandatory compliance obligations**.
By SGS certifications these framework requirements are met.

Layer 2 – Governance & Management System



Area	Standards	Role
Functional Safety	• ISO26262 • IEC 61508	Functional Safety Management
Cybersecurity	• IEC 62443	Information security management (if data processing involved)
AI Safety	• ISO PAS 8800	AI Management System (AIMS) governance, risk processes, documentation, monitoring
	• ISO/IEC23894 • ISO CD 22440	AI risk management guidance



Organizational-level control of AI lifecycle, risk tracking, accountability.

Layer 3 – Machinery & Robotics Safety



Area	Standard	Role
Safety	• ISO 12100	Risk assessment methodology
	• ISO 10218	Robot Basic Standard
	• ISO 13849	Performance Levels (PL) for safety functions
	• ISO/TS 15066	Robot Collaboration
	• ISO 25785*	Mobile Robots
	• ISO 13482	Safety requirements for personal care robots

*not yet published

Layer 4 – System Level of development



Area	Standards	Role
Functional Safety	• IEC 61508 • ISO 26262 • ISO 13849	System Safety requirements and verification
SOTIF*	• ISO 21448	Safety Requirements to the intended functionality
Cybersecurity	• IEC 62443	System Security requirements
AI	• ISO PAS 8800	Requirements for safe AI Artificial intelligence
	• ISO/IEC TR 5469:2024 • ISO 22440-1/2/3	Functional safety and AI systems

*SOTIF-for-Robotics–Further Development, basic concept of ISO 21448 transferred

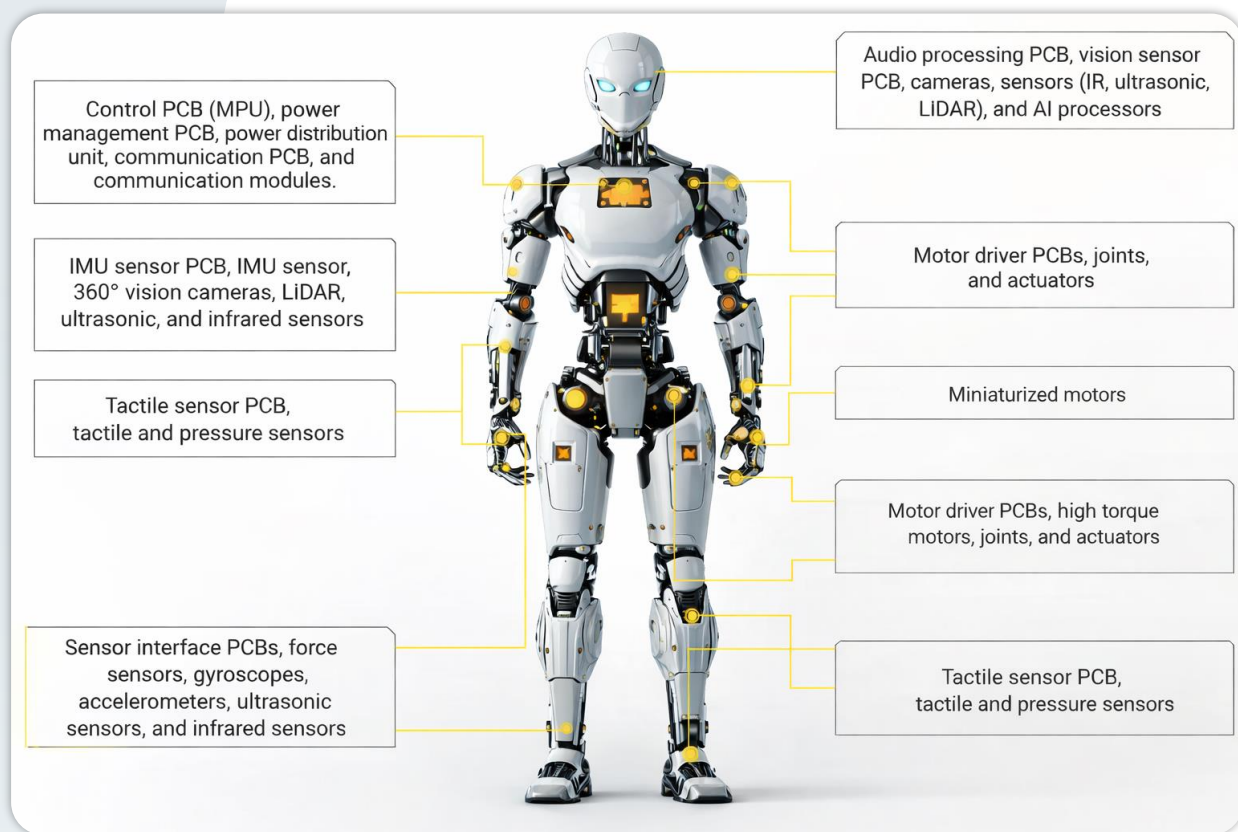
Layer 5 – Hardware- and Software Level of development



Area	Standard	Role
Functional Safety	• IEC 61508 • ISO 26262 • ISO 13849	Functional Safety on Hardware and Software
	• ISO 13482	Design and safety requirements for semiconductor (human interaction with robots)
Cybersecurity	• IEC 62443	Cybersecurity on Semiconductor level
AI	• ISO PAS 8800	Safety and Artificial Intelligence
	• ISO 22440	AI risk management on Semiconductor

第5层 – 硬件/单色 - 半导体（示例）

人工智能类机器人中半导体组件的分配



所有子系统均基于或包含先进半导体技术。

半导体技术的应用包括：

- 从人工智能类机器人环境（摄像头、激光雷达、红外传感器等）收集数据的感测层
- 感知与决策层 解读传感器数据并做出决策。（NPU处理器，GPU处理器.....）
- 控制/执行层将决策转化为物理操作。（关节和执行器,...）

因此，人工智能类机器人中使用的每一种半导体都至关重要

2026欧美合规落地路径



人形机器人认证框架(商用/家用)

产品类型 Appliance	认证类型 Type of Certificate	相关标准 Relative Standards	
 轮式人形机器人	- 机械指令 - CE-MD 2006/42/EC *低电压指令 - CE-LVD 2014/35/EU - 整机发CE证书前提是满足功能安全	- ISO 13482 - EN ISO 3691-4/ISO 10218	个人助理机器人专有安全标准，包含 酒店送餐机器人，外骨骼机器人，载人服务机器人，配送机器人等工业用
		- EN 60335-1 (家用环境或商用) - EN 1175/EN 60204-1	电气安全 工业用
		EN ISO 12100	风险评估
	电磁兼容指令CE-EMC 2014/30/EU	- EN 61000-6-1 & EN 61000-6-3 - EN 55014-1/2 & EN 61000-3-3/-2	商业及轻工业环境 家用环境
	无线指令CE-RED 2014/53/EU	WIFI, RFID, Bluetooth...	EN 300系列，适用标准取决于频率
	- ROHS指令 - CE-ROHS 2011/65/EU	Depends on material	民用或家用需要考虑
	功能安全Functional Safety	EN ISO 13849-1 (可选)	工业整机功能安全【参照ISO 13482表一】
	软件评估Software Evaluation	EN IEC 60730-1 (可选)	家用或者商用可选
	网络安全Cyber Security	PrEN 50742/EN 62443	网络安全评估
	- 北美市场 - NA Market	- FCC - UL 3300 - UL3100+ANSI15.06	强制 产品安全标准

人形机器人认证框架(商用/家用)

产品类型 Appliance	认证类型 Type of Certificate	相关标准 Relative Standards	
 双足人形机器人	- 机械指令 - CE-MD 2006/42/EC *低电压指令 - CE-LVD 2014/35/EU 整机发CE证书前提是满足功能安全	- ISO 13482 - ISO CD 25785-1 (工业)	个人助理机器人专有安全标准 (双足人形机器人)
		- EN 60335-1 (家用环境或商用) - EN 1175/EN 60204-1	电气安全 工业用
		EN ISO 12100	风险评估
	电磁兼容指令CE-EMC 2014/30/EU	- EN 61000-6-1 & EN 61000-6-3 - EN 55014-1/2 & EN 61000-3-3/-2	商业及轻工业环境 家用环境
	无线指令CE-RED 2014/53/EU	- WIFI, RFID, Bluetooth... - Depends on frequency	EN 300系列, 适用标准取决于频率
	- ROHS指令 - CE-ROHS 2011/65/EU	Depends on material	民用或家用需要考虑
	功能安全Functional Safety	EN ISO 13849-1 (可选)	工业整机功能安全【参照ISO 13482表一】
	软件评估Software Evaluation	EN IEC 60730-1 (可选)	家用或者商用可选
	网络安全Cyber Security	PrEN 50742/EN 62443	网络安全评估
	- 北美市场 - NA Market	FCC	强制
		- UL 3300 - UL3100+ANSI15.06	产品安全标准

复合机器人认证框架(工业环境)

产品类型 Appliance	认证类型 Type of Certificate	相关标准 Relative Standards	
 复合机器人	- 机械指令/法规 - CE-MD 2006/42/EC - CE-MR EU 2023/1230 *低电压指令 - CE-LVD 2014/35/EU	- ISO 3691-4 (Product safety standards) - ISO 10218-1(Product safety standards)	(轮式人形机器人工业场景应用可参照此方案)
		- EN 1175 (Electrical Safety) - EN 60204-1 (Reference)	电气安全
		EN ISO 12100	风险评估 (A类)
	- 电磁兼容指令 - CE-EMC 2014/30/EU	- EN 61000-6-2 & EN 61000-6-4 - EN 12895	工业环境
	- 无线指令 - CE-RED 2014/53/EU	- WIFI, RFID, Bluetooth... - Depends on frequency	EN 300系列, 适用标准取决于频率
	功能安全Functional Safety	ISO 13849-1	整机功能安全
	网络安全Cyber Security	PrEN 50742/EN 62443	网络安全评估
	NA Market	FCC – SDoC (USA)	强制标准
		- ANSI/RIA R15.08 or - UL1740+ANSI/RIA R15.06	AMR安全标准



商用清洁机器人认证框架

产品类型 Appliance	认证类型 Type of Certificate	相关标准 Relative Standards	
 整机部分 Complete Machine	- 机械指令 - CE-MD 2006/42/EC - 低电压指令 - CE-LVD 2014/35/EU	- EN ISO 12100(Risk Assessment) - Annex 1 of 2006/42/EC	整机发CE证书前提是满足功能安全
		- EN IEC 60335-1 - EN IEC 60335-2-72 &Test according to EN IEC 63327	
	- 电磁兼容指令 - CE-EMC 2014/30/EU	EN 61000-6-1 & EN 61000-6-3	商业及轻工业环境
	- 无线指令 - CE-RED 2014/53/EU	- WIFI, RFID, Bluetooth... - Depends on frequency	适用标准取决于频率
	- ROHS指令 - CE-ROHS 2011/65/EU	Depends on material	可以考虑
	- 功能安全 - Functional Safety	EN ISO 13849-1	整机功能安全【安全功能参照IEC 63327表一】
	- 北美市场 - NA Market	- FCC - UL 60335-1 - UL 60335-2-72 - CSA/ANSI C22.2 No. 336-17	强制 产品安全标准 (功能安全部分可以直接用欧标部分，无需二次评估，美标功能安全也是用ISO 13849-1评估)

关键零部件与认证



传感器

CE(LVD/RED/EMC) | NRTL
EN/IEC/UL 61010-1
EN/IEC/UL 61496



电池

CE(LVD) | NRTL
EN/IEC 62619
EN/IEC 62133
UL 2271/UL 2054/UL 2580



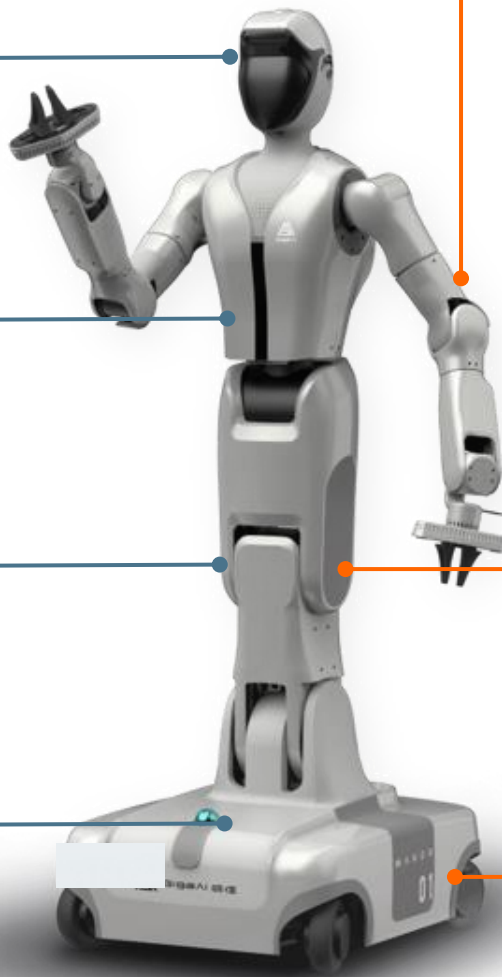
电机/驱动

CE(MD,LVD,EMC) | NRTL
EN/IEC/UL 61800-5-1
EN/IEC 60034-1
UL 1004



关节模组

CE(LVD/EMC) | NRTL
EN/IEC/UL 61800-5-1/-5-2/-3



机器人系统/灵巧手

CE(MD,LVD,EMC) | NRTL
EN/ISO 10218-1/2
EN/ISO 13482
EN/ISO 15066
EN/IEC 60204-1
EN/IEC 60335-1
EN/IEC 62368-1
NRTL ANSI/RIA R15.06
NRTL ANSI/RIA R15.08
UL3100
UL 3300
ISO/CD 25785-1



充电器

EN 60335-2-29
Or EN 61558-2-16
Or EN 62477
UL 1973
UL 1564



控制器/无线模块

EN 300系列
EN 301系列
EN 62311

SGS认证项目整体流程



← **Listed Certificate**

LPC Label



- 2.0-DoC & Identity of Products
- 3.0-Electrical Diagram & BOM-OK
- 4.0-Pneumatic Diagram & BOM-Ok
- 5.0-Components Certs-OK**
- 6.0-Overall Drawings & Guard Drawing-OK
- 7.0-Manual & Technical data
- 8.0-Nameplate & Warning label-Ok
- 9.0-Machine Photos-OK
- 10.0-ISO 9001 Cert-Ok

签发证书

提交全套技术资料及
报告给SGS签证官签
发证书



项目立项

项目开始，确定
SGS与客户技术对
接人，启动认证项
目

图纸审核

审核图纸等技术资
料是否与标准架构
相同，材料选用是
否合规

现场/实验室测试

样机完成后，SGS
会对样品进行测试，
确定样品是否符合
标准要求，是否需
要整改

资料准备

客户准备产品技术资
料，SGS编写测试报
告，进行发证前的准
备【美标需要验厂】



合规案例:



合规认证

CE, NRTL, 功能安全, 网络安全, 全球市场准入



性能测试

环境测试, 可靠性测试, 稳定性测试



培训

法规解读, 标准培训



思灵Agile ONE 人形机器人

- **产品特性:** 物理AI, 工业场景设计.
- **应用场景:** 自动化, 3C电子产线.



云深处“Lite3” 四足机器人

- **产品特性:** 防尘防水等级 66 级, 自主巡检
- **应用场景:** 电力、消防安全、工业场景



无界动力KITT15轮式底盘人形机器人

- **产品特性:** MD指令+功能安全评估.
- **应用场景:** 自动化, 3C电子产线.

证书样本

SGS

VERIFICATION OF MD COMPLIANCE

No.:

MD SHES26

Applicant:

Anj
Rooms 705, 706, 712, 713, 714, 719, 720, 7th Floor, Building 2,
Courtyard 16, East Bank Road of Xiaoyue River, Haidian District,
Beijing, China

Manufacturer:

Anj
Rooms 705, 706, 712, 713, 714, 719, 720, 7th Floor, Building 2,
Courtyard 16, East Bank Road of Xiaoyue River, Haidian District,
Beijing, China

Product Name:

Semi-Humanoid Robots

Model No.:

KI

Trademark:



Additional information (if any):

Sufficient samples of the product have been tested and found to be in conformity with requirements of following standard(s) and applicable parts of the Annex I of the Machinery Directive (2006/42/EC).

Test Standard:

EN 60204-1:2018+A1:2025
EN ISO 12100:2010
ISO 13849-1:2023
ISO 13849-2:2012

as shown in the

SHE5260400749401-01/02

Test Report Number(s):

SHE5260400031971

This Verification of MD Compliance has been granted to the applicant based on the results of tests, performed by Laboratory of SGS-CSTC Standards Technical Services Co., Ltd. on sample of the above-mentioned product in accordance with the provisions of the relevant harmonized standards under the Machinery Directive 2006/42/EC. The CE mark can be affixed, under the responsibility of the manufacturer, after completion of an EC Declaration of Conformity and compliance with all relevant EC Directives. The affixing of the CE marking presumes in addition that the conditions in the Directives are fulfilled.

Andrew Zhai

Technical Manager

SGS-CSTC Standards Technical Services Co., Ltd.

2026-07-13



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Member of the SGS Group (806 84)

Safety-VOC-P22 Rev.2.1 2024-07-18

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SGS

Functional Safety

Inspection Certificate No. FS/SHFS2

Applicant:

o., Ltd.
Rooms 705, 706, 712, 713, 714, 719, 720, 7th Floor, Building 2,
Courtyard 16, East Bank Road of Xiaoyue River, Haidian District,
Beijing, China

Manufacturer:

o., Ltd.
Rooms 705, 706, 712, 713, 714, 719, 720, 7th Floor, Building 2,
Courtyard 16, East Bank Road of Xiaoyue River, Haidian District,
Beijing, China

Trademarks:

Product Name:

Semi-Humanoid Robot

Model (s) No. /

Series:

Assessment

ISO 13849-1:2023, ISO 13849-2:2012

Performed:

Conclusion:

The safety architecture and performance level meet PL d with
Cat. 3 according to ISO 13849-1:2023, detail information of
safety function items is shown in APPENDIX.

As shown in the technical report number(s):

SHFS260400031971

This inspection certificate confirms the achievement of the requirements of functional safety based on proof of the safety-related parameters (failure rate, DC, PFH, safety architecture etc), proofs that processes, and methods are established at the manufacturer guaranteeing that unexceptionable processes in terms of risk analysis, design, production, validation, modification and quality management comply with the standard.

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Approved by:

Issue Date:

02/07/2026



Andrew Zhai

Technical Manager

Expiry Date:

01/07/2031

SGS-CSTC Standards Technical Services (Shanghai) Co., Ltd.

No 588, West Jindu Road, Songjiang District, Shanghai, 201612.

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SGS

Functional Safety

Certificate No. FS/SHFS

Certificate Holder:

ny) GmbH
lünchen, Germany

Manufacturer:

ny) GmbH
lünchen, Germany

Trademarks:

Certified Product:

Humanoid Robot P2-1L, Humanoid Robot P2.1,
Humanoid Robot P2.0 Arm, Humanoid Robot P2.1 Arm

Model (s) No. /

Arm

Series:

Assessment

ISO 13849-1:2023, ISO 13849-2:2012, IEC 62061:2021

Performed:

Conclusion:

The safety architecture and performance level meet PL d with
Cat. 3 according to ISO 13849-1, and SIL 2 with HFT-1 (1oo2)
according to IEC 62061, detail information of safety function
items is shown in APPENDIX.

As shown in the technical report number(s):

SH-
SH-

This certificate confirms the achievement of the requirements of functional safety based on proof of the safety-related parameters (failure rate, DC / PFH, safety architecture etc), proofs that processes, and methods are established at the manufacturer guaranteeing that unexceptionable processes in terms of risk analysis, design, production, validation, modification and quality management comply with the standard.

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Authorized by:

Effective date:

07/03/2025



Andrew Zhai

Technical Manager

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